University of Cape Town

University Examinations – May/June 2019

Department of Mathematics and Applied Mathematics

MAM3000W

Module 3MS: Metric Spaces

Time : 2 hours **Full Marks: 40**

Marks available: 40

Please answer Question 1 in one answer book, and Questions 2 to 5 in another.

1. Suppose that (X, d_X) and (Y, d_Y) are metric spaces and $f : X \to Y$ is a function. For $a, b \in X$, define

$$p(a, b) = d_X(a, b) + d_Y(f(a), f(b))$$

   (a) Prove carefully that p is a metric on X.

   (b) Write down the definition of the diameter of a subset of a metric space.

   (c) Now let

   - $(X, d_X) = (\mathbb{R}, d_D)$ where d_D denotes the discrete metric
   - $(Y, d_Y) = (\mathbb{R}, d_E)$ where d_E denotes the Euclidean metric
   - $f(x) = x^2$

   and define p as described above. In the metric space $(\mathbb{R}, p)$:

   i. Find all real numbers in the open ball $B(\sqrt{26}; 11)$. Show brief working.

   ii. Find the diameter of the interval $[-4, 4]$. (No working required.)

[10]

Please change to a new answer book now.

2. Suppose that X and Y are metric spaces, and $f : X \to Y$ is a function.

   (a) Prove that if f is continuous, then, for all $A \subseteq X$, $f(\overline{A}) \subseteq \overline{f(A)}$.

   (b) Prove or disprove:
   f is continuous if and only if, for all $B \subseteq Y$, $\overline{f^{-1}(B)} = f^{-1}(\overline{B})$.

[6]

3. Read the following theorem and proof, and then answer the questions that follow.

Theorem Suppose that (X, d) is a metric space and A is a subset of X. If A is a complete subset of X, then A is a closed subset of X.

Proof Suppose that A is complete. (1)
Take a sequence (x_n) such that $x_n \to x$ in X and $x_n \in A$ for all n. (2)
Then (x_n) is a Cauchy sequence. (3)
So $x_n \to a$ for some $a \in A$. (4)
So $x = a$, since limits are unique. (5)
So $x \in A$. (6)

(a) Prove that every convergent sequence in (X, d) is Cauchy, as used in line (3).

(b) How do we know that $x_n \to a$ for some $a \in A$, as claimed in line (4)?

(c) Explain what is meant by the expression "limits are unique" (in line (5)) and prove it.

(d) Is the converse of this theorem true or false? Prove it or disprove it. [9]

4. (a) i. Give brief reasons why, in any metric space, $B(a;r) \subseteq \text{int}B[a;r]$.
   ii. Give an instance where $B(a;r) \neq \text{int}B[a;r]$.

(b) Prove that every compact metric space is bounded.

(c) Prove or disprove: If (X, d_X) and (Y, d_Y) are connected metric spaces, and $X \times Y$ has a metric ρ that induces componentwise convergence, then $(X \times Y, \rho)$ is connected.

[12]

5. (a) For any metric space X, prove that there exists a metric space Y and an isometry $f : X \to Y$ such that $f(X)$ is not an open subset of Y.

(b) Prove or disprove: If (X, d) is a compact metric space and $f : (X, d) \to (X, d)$ is an isometry, then f is onto. [3]

THE END